Data Driven Control Integration Week

Group 8 (MPC)

Christian Brunet Jules Marceau
Matthieu Moussaoui Clément Tamboise
Rayane Trigoust

CentraleSupélec

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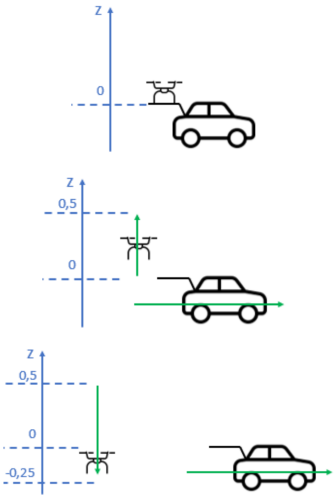
Project Objectives

Available Resources:

- Dataset of vertical acceleration.
- MATLAB/Simulink drone simulator.

Objectives:

- Validate the dataset.
- Identify the model.
- Develop a control law.



Dataset Validation

Cleaning the data

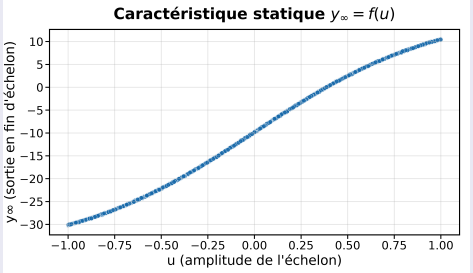
Remove lines that includes:

- Frozen sensor for 2+ seconds
- 5+ diverging values

Normalisation

- Directly in the training

Static characteristic



$y_\infty = f(u)$ after eliminating corrupted runs.

System Modeling

Linear Model

State Function

First-order lag driven by $N(u)$:

$$\begin{cases} \dot{x}(t) = -\dfrac{x(t)}{\tau} + \dfrac{N(u(t))}{\tau} \\ a_z(t) = x(t) \end{cases}$$

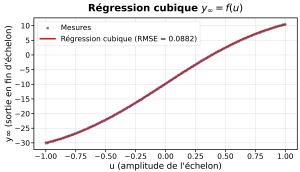
Parameter Identification

- **Static characteristic** $N(u)$: 3rd-order polynomial identified on steady-state regimes:

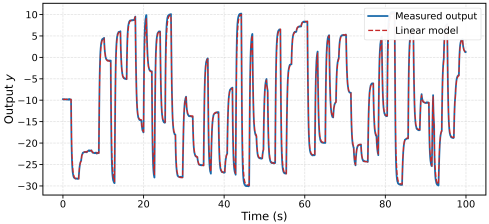
$$a_z = N(u) = a_3 u^3 + a_2 u^2 + a_1 u + a_0$$

- **Tau identification**

$a_3 \approx -6.05,\ a_2 \approx 1.1 \times 10^{-3},\ a_1 \approx 26.1,\ a_0 \approx -9.81,\ \tau \approx 0.22$ s



Linear Model Validation



First RNN

Sequence-to-sequence architecture

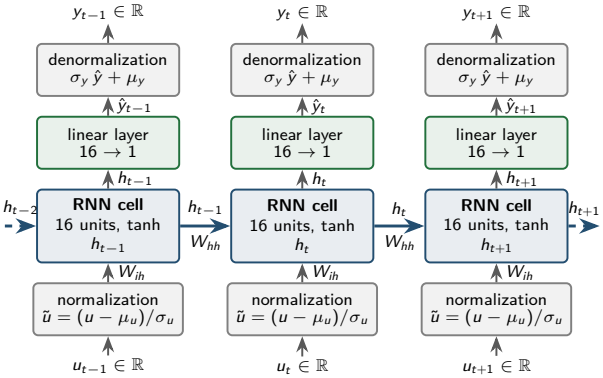
Parameters

- One scalar input u_t and one scalar output $\hat{y}_t$
- Hidden state dimension: 16
- Early stopping to avoid overfitting

State representation

$$h_t = \tanh(W_{ih} u_t + W_{hh} h_{t-1} + b)$$

$$\hat{y}_t = W_{hy} h_t + b_y$$



u: PRBS input $y = a_z$: vertical acceleration step $T_e = 0.05$ s

First RNN

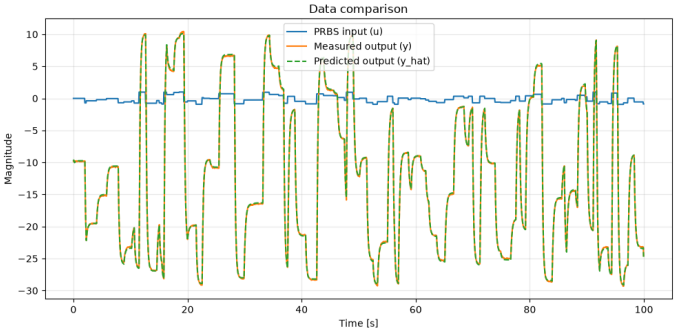
Results

Numerical results

- Validation loss: 3.93×10^{-4}
- Test loss: 4.01×10^{-4}

Training criterion

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2$$



Z-axis control

Designing the reference trajectory

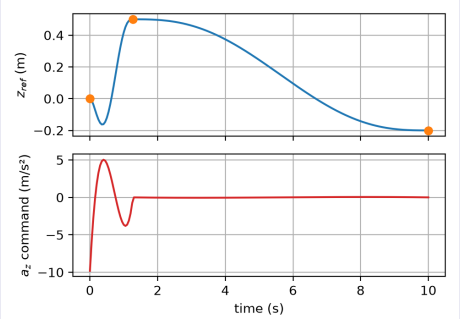
Principle

- **Input**: waypoints z_i and passage times t_i
- One **polynomial function** per segment $[t_i, t_{i+1}]$, imposing z, $\dot{z}$ and $\ddot{z}$ at both ends
- Null speed at waypoints, null acceleration on arrival, $\ddot{z}$ continuous
- **Output**: $z_{ref}(t)$ and the acceleration command $\ddot{z}_{ref}(t)$

Results

Smooth analytic reference; overshoot beyond waypoints removed by shortening segments

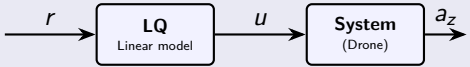
Example: $z = 0, 0.5, -0.2$ m at $t = 0, 5, 10$ s



LQ control

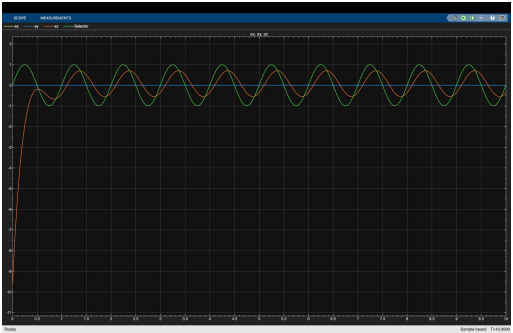
Linear-quadratic control design

Control principle



Performance

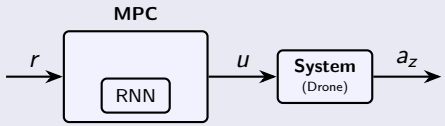
- Global trajectory following
- Wrong amplitude + phase shift
- Missing prediction



MPC control

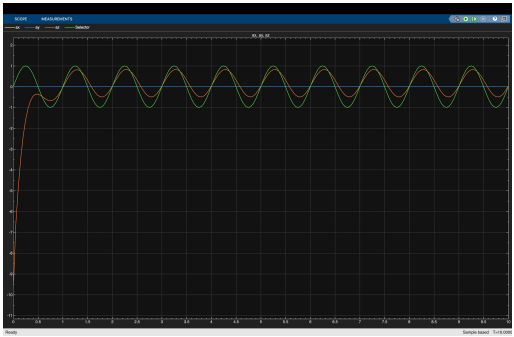
Model predictive control design

Control principle



Performance

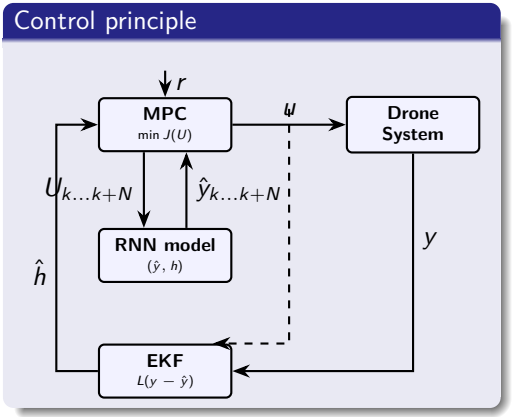
- Good overall tracking.
- No phase shifting good prediction.
- Full amplitude not reached.
- Only internal state.

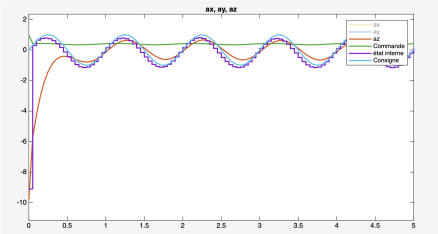


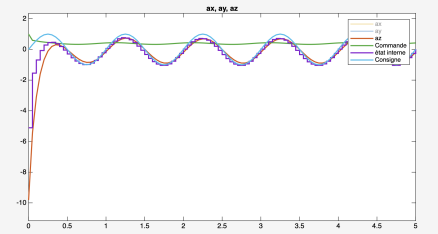
MPC with EKF

State estimation and predictive control

Control principle

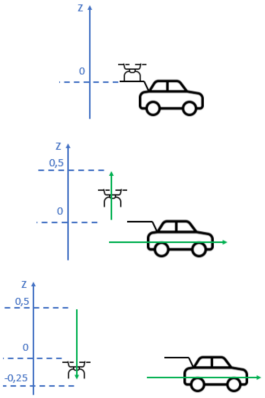


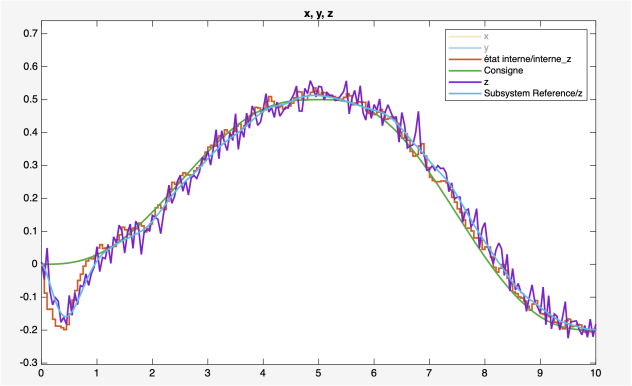




Requested command

MPC control with EKF feedback





Comparison

LQ, MPC and MPC with EKF

LQ control with Linear Model

- Simple and very fast.
- Lack prediction.
- Phase shifting.

MPC control with RNN

- Prediction.
- Internal state diverges from reality.
- No feedback loop.

MPC with EKF

- Feedback from measurements.
- Position control.

Data generation

PRBS excitation in the 3D drone workspace

Objective

Generate sufficiently rich input/output trajectories to identify a nonlinear dynamic model of the drone.

- Excite the four actuator commands around hover
- Capture the coupled 3D dynamics (12 variables)
- Store synchronized inputs and measured outputs

Simulation setup

- Sampling time: $T_s = 0.05\,\mathrm{s}$
- $N_{\text{traj}} = 2000$ samples per trajectory

APRBS modes

We don't directly generate u inputs (unusable data). Instead, we use a flight controller : the inputs are now four independently generated piecewise-constant modes:

$$\delta = [\delta_T, \delta_r, \delta_p, \delta_y]$$

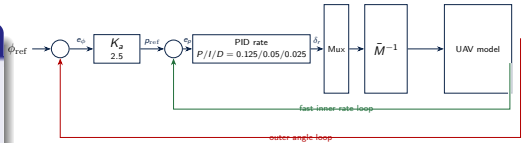
Each level is sampled uniformly and held for a random duration between 1 and 3 seconds.

Data generation

Flight controller

From version 1 to version 2

The first version used PID controllers directly on the attitude angles. This solution was stable, but not fast enough for the drone dynamics (see appendix).
Version 2 therefore uses a cascade controller: an outer angle loop generates an angular-rate reference, followed by a fast inner PID loop based on the gyroscope measurement.



Controller parameters

- Outer angle gain: $K_a = 2.5$
- Inner rate PID:
  $P = 0.125,\ I = 0.05,\ D = 0.025,\ N = 100$

Appendix: System Modeling

Linear Model: Least Square Method

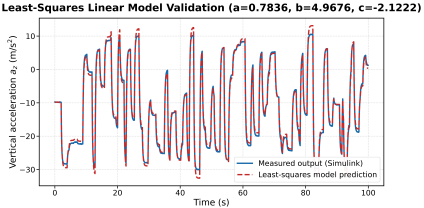
Model equations

$$y_k(t+1) = a\,y_k(t) + b\,u_k(t) + c$$
$$= \varphi_k^{\mathsf{T}}\theta.$$

$$\varphi_k^{\mathsf{T}} = \begin{bmatrix} y_k(t) & u_k(t) & 1 \end{bmatrix} \qquad \theta = \begin{bmatrix} a & b & c \end{bmatrix}^{\mathsf{T}}.$$

The least-squares estimate is

$$\theta^{\star} = \left(\mathbf{\Phi}^{\mathsf{T}}\mathbf{\Phi}\right)^{-1}\mathbf{\Phi}^{\mathsf{T}}\boldsymbol{Y}.$$



Appendix: System Modeling

Linear Model with Constant Steady-State Gain

State Function

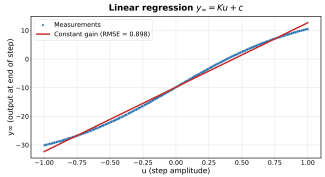
First-order lag driven by a constant gain K and offset c: $\dot{v}(t) = \frac{1}{\tau}\big(Ku(t) - v(t)\big)$, i.e. $\tau\,\dot{v} + v = Ku + c$.

Parameter Identification

- **Static gain** K: affine least-squares fit on steady-state regimes:

$$v(t) = K\,u(t) + c, \qquad K \approx 22.5,\ c \approx -9.8$$

- **Tau identification**



Linear Model Validation

